

Tension by Restriction: Designing Strategic Gameplay Through Intentional Constraint in Pet Simulation

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Research Focus Statement

This study explores how mechanical constraints (e.g., restricted camera views, limited interventions, and delayed action effects) can be intentionally used in simulation games to create psychological tension and promote strategic decision-making. Set in a remote pet monitoring scenario, the game requires players to manage an active, disruptive pet using limited surveillance tools and teleoperated gadgets, and react selectively to prevent harm. Unlike traditional pet simulation games that emphasize relaxation and emotional connection, this project transforms passive observation into an active, tension-driven strategy system. By embedding systemic constraints into core mechanics, the game demonstrates how constraints can focus attention, guide players to play more strategically, and increase engagement without relying on punishment or moral consequences.

Research Gap and Evidence

Despite strong commercial growth, virtual pet simulators rarely use constraint-driven mechanics. This study, conducted by Sensor Tower in 2024, evaluated two design elements which are mechanical constraints (e.g., limited monitoring range, limited interaction, or delayed feedback) and stressful mechanics (e.g., resource scarcity, risk of failure, or strategic ambiguity).

The analysis showed that 18 of the 20 games relied on continuous control loops (e.g., feeding, customization, or care) with continuous visibility and immediate feedback. None of the games implemented systematic constraints, such as limited view angles, intervention caps, or action delays. The remaining two games only used very few restrictive mechanisms. The game Happy Pet Story has intervention cooldown constraints, and some tools require waiting for several minutes before they can be used again. And Talking Ben has a limited number of "punishment" mechanisms, and the number of times it can intervene in bad pet behavior is capped. The former serves time rhythm management and payment guidance, while the latter is to enhance character performance and fun. Neither of them recreates strategic decision-making pressure or cognitive challenges. Overall, they are still relaxing games without strategy.

According to Research and Markets (2025), the global virtual pet simulator app market is expected to grow from \$326.14 million in 2025 to \$525.05 million in 2030, reflecting high user demand and market saturation. The study mentions that "strategy-oriented simulators emphasize progression mechanisms that reward careful planning," but does not mention commercial precedents for restriction mechanisms that induce tension and increase the strategic nature of gameplay. This indicates a clear market and design gap: while the genre is booming in terms of revenue and user base, its gameplay remains largely

unrestricted. This study fills this gap by intentionally embedding restriction mechanisms into the core design, transforming pet simulation into a field of conscious cognitive challenge and strategic gameplay.

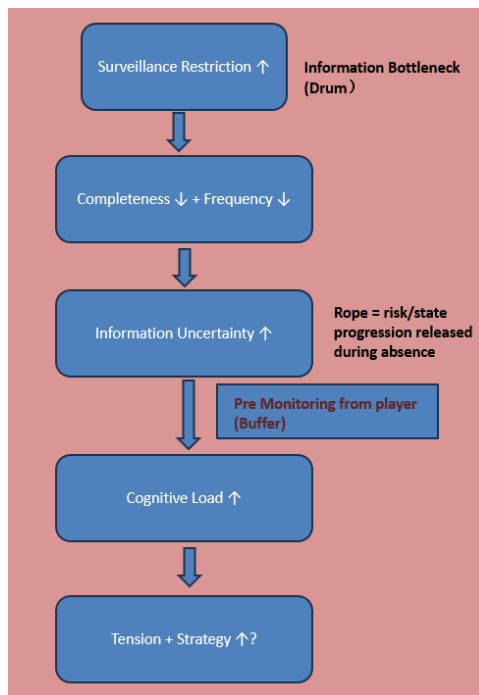
Theoretical Framework-TOC

This research is informed by principles derived from the Theory of Constraints (TOC), which proposes that system behavior is primarily shaped by its main limiting factor. Rather than optimizing all system components equally, TOC emphasizes identifying the primary constraint that governs overall system performance.

In this study, the primary system constraint is monitoring time bandwidth, defined as the amount of time players can spend observing surveillance information. Instead of directly limiting player abilities or action mechanics, the system restricts access to real-time environmental information by requiring players to allocate time to work tasks. As work occupancy increases, available monitoring bandwidth decreases.

This reduction in monitoring time bandwidth increases informational uncertainty. Players must make decisions based on incomplete or delayed knowledge of system states. Within this framework, Work-In-Progress (WIP) is reinterpreted as cognitive and informational workload—the number of system states players must mentally track while away from surveillance. As monitoring interruptions increase, WIP accumulates cognitively rather than physically.

The conceptual model can therefore be expressed as a causal chain:



By manipulating monitoring time restriction, the system regulates informational throughput, cognitive pressure, and ultimately strategic adaptation. Surveillance restriction functions not as a deprivation of agency, but as a structural bottleneck that reorganizes player behavior under uncertainty.

Literature Review

1. . “A constraint limits the output of every system, whether we acknowledge it or not. When properly identified and managed, **constraints provide the fastest route to significant improvement** and form the bedrock for continuous growth. When ignored, the constraint may lie idle, squandering system capacity. An out-of-control constraint may also wreck havoc on delivery schedules and cause unpredictable delays.” .

According to the TOC Institute, “a constraint limits the output of every system, whether we acknowledge it or not. When properly identified and managed, constraints provide the fastest route to significant improvement and form the bedrock for continuous growth. When ignored, the constraint may lie idle, squandering system capacity. An out-of-control constraint may also wreck havoc on delivery schedules and cause unpredictable delays” (TOC Institute, n.d.).

This perspective reframes constraints not as weaknesses to be eliminated, but as structural determinants of system performance. Within interactive systems, such as games, this logic suggests that a deliberately designed bottleneck can reorganize player behavior and redirect strategic focus. In this study, surveillance restriction functions as the primary constraint that governs informational throughput. By structuring when and how players access monitoring information, the system reorganizes decision-making behavior under uncertainty.

2. Finn Atkinson and Laura Pruszko. 2025. Player Agency Under Constraint: A Pilot Study on the ForcedChoice Effect in Narrative Difficulty Design. In Proceedings of the 2025 Conference on Creativity and Cognition (C&C '25). Association for Computing Machinery, New York, NY, USA, 404–408.
<https://doi.org/10.1145/3698061.3735326>

This article describes a “forced-choice effect” observed in their study of constrained decisionmaking in Elden Ring, where players, when faced with limited options, tended to focus on dominant strategies. They note that “players converged toward dominant strategies—with variation in how they interpreted them.” In my game, similar constraints—such as limited camera access and intervention chances—push players to prioritize actions and develop adaptive strategies under pressure. This supports the idea that intentional mechanical limitation can sharpen decisionmaking and increase psychological engagement, even when freedom is reduced.

3. Spiliopoulos, L., Ortmann, A., & Zhang, L. (2018). Complexity, attention, and choice in games under time constraints: A process analysis. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 44(10), 1609–1640. <https://doi.org/10.1037/xlm0000535>

This article shows that when players face time pressure, they tend to focus attention, simplify decision-making, and rely on quicker strategies without reducing performance. In my game, players are not under strict time pressure but instead face limited spatial and tool-based agency. The fixed number of cameras and restricted use of intervention tools (like toys or distractions) means players must selectively attend to the most relevant cues and prioritize their actions

carefully. This aligns with Spiliopoulos et al.'s observation that constraint reshapes player cognition, encouraging deliberate, strategic action and heightened engagement even when temporal urgency is absent.

Historical-Technical Survey

1. Five Nights at Freddy's (Scott Cawthon, 2014)

This game relies on static surveillance cameras, limited energy, and delayed feedback on actions to generate stress and fear. Players must balance limited resources with fragmented visual information to survive the nighttime threats. Its effectiveness comes from forcing players to make decisions with incomplete data and time-sensitive constraints.

My game borrows from its surveillance interface and constraint mechanisms, such as limited visibility and limited intervention, but removes terror and punishment as incentive structures, applying these same systems to a care setting, transforming fear into focused engagement. The game retains the strategic tension of FNAF, but replaces fear with cognitive urgency, making it suitable for non-violent simulation while maintaining emotional intensity.

2. Talking Ben (Outfit7, 2011)

Talking Ben is part of the Talking Tom series and is primarily designed as a casual entertainment app. While most interactions are open-ended and humorous, the game introduces a limited use "scold" feature that allows the player to briefly interrupt Ben's behavior a few times a day. This mechanic implies the scarcity of intervention but is designed to build in comedic timing rather than challenge the player's decision-making.

My game learns from this idea of limiting player interaction, but recontextualizes it as a strategic decision-making system rather than a novelty. The limiting mechanic that builds emotional feedback is replaced by a delayed, meaningful intervention system that pushes the player to think strategically, making each action cognitively meaningful rather than pure entertainment.

3. Tamagotchi (Bandai, 1996)

Tamagotchi pioneered virtual pet care by providing a minimalist interface with only a few options to choose from. Despite its simplicity, the system creates a deep emotional attachment through lasting consequences - if the user ignores their pet, it may become sick or die. This is one of the first digital examples of limited control generating a large psychological investment.

My project takes this principle of “consequences for inaction” and expands on it by placing the player in the role of a surveillance agent with incomplete information, increasing tension through visual constraints and time delays.

Artifact Pre-Production

This thesis explores how constrained mechanical systems (e.g., limited monitoring, delayed actions, and limited interventions) can create psychological tension and cognitive engagement in a remote pet monitoring simulation. Built for PC using Unreal Engine 5, this prototype improves the playability and strategy of the casual pet simulation genre through a strategic decision-making perspective based on partial information.

The game puts players in the role of a pet owner who is away from home for work, monitoring their pets through a network of fixed cameras accessed through a smartphone interface. Gameplay revolves around switching camera views, detecting destructive behavior (e.g., knocking over objects, chewing furniture), and responding with limited interventions (e.g., voice commands or virtual toys). Interventions are controlled by cooldowns and delays, forcing the player to think carefully and prioritize.

A pet status interface is displayed before the monitoring phase begins. The interface displays key indicators of the pet, such as mood, hunger, and destructive drive. These values are updated over time and affect the likelihood of the pet to engage in destructive behavior. The player must interpret these indicators and determine in advance when to intervene or even switch cameras. This mechanism adds a layer of prediction and resource management to the system, encouraging the player to combine real-time monitoring with strategic foresight.

Unlike traditional pet simulators, this system intentionally deprives players of full autonomy. The goal is not constant interaction, but strategic inaction and anticipation. In "Thesis I", the pet's behavior is driven by a random script, but in Thesis II this could be extended to a finite state machine or lightweight AI. The emphasis is on the player's ability to manage uncertainty and allocate limited attention, especially in the late stages of introducing background distractions - the situation of going to work.

Design methods

Content analysis: A systematic review of 20 commercially successful mobile pet simulators

(Table1,2025) observed and studied industry trends towards unrestricted interaction and immediate feedback. These findings highlighted a gap in the research of constraint-based systems in the genre and guided our decision to pursue a set of mechanisms to improve cognitive levels. Theory-driven framework: The design draws on the MDA framework to

structure how mechanical constraints (e.g., fixed viewpoint and input delay) create dynamic changes in priority, tension, and focus. The theory supports the core hypothesis that constraints rather than freedom enhance psychological immersion.

Game Title	Platform	Main Mechanics	Constraint Mechanics Present?	Notable Emotional Engagement
My Talking Tom	Mobile	Care, Feed, Dress Up	No	High - Pet animations & feedback
Pou	Mobile	Feed, Clean, Mini-Games	No	Moderate
Bubbu	Mobile	Pet Care, Decorate	No	Moderate
My Talking Angela	Mobile	Care, Fashion, Makeup	No	High - Expressive style
Happy Pet Story	Mobile	Pet Simulation, Decor	Yes (Cooldowns)	High- Whimsical tone
Pet Idle	Mobile	Care, Leveling, Build Rooms	Yes (Resource Wait)	Moderate- Routine Loop
Cat Condo	Mobile	Idle Merge, Collection	Yes (Merge Limits)	Low -Idle Design
Dog Town	Mobile	Care, Breed, Upgrade	Yes (Time for Training)	Low
Hamster Life	Mobile	Feed, Clean, Watch	No	Moderate- Pet Sounds
Pet Shop Story	Mobile	Decorate, Breed	Yes (Room Unlock Timers)	Moderate
Moy	Mobile	Simple Care Loop	No	Low
Talking Ben	Mobile	Talk, Interact	Yes (Limited Voice Use)	High- Voice response
Pet World	Mobile	Animal Care, Rescue	Yes (Care Slots)	Moderate
Pakka Pets	Mobile	Care, Grow Creatures	No	Moderate
Clawbert	Mobile	Collect Claw Toys	Yes (Random Reward System)	Low- Gacha Tone
Cat Room	Mobile	Decorate, Feed	No	Moderate
Pet Hotel	Mobile	Manage Hotel for Pets	Yes (Room Full)	Moderate
My Virtual Pet	Mobile	Mini-Games, Basic Care	No	Low
Pet Rescue Saga	Mobile	Puzzle + Rescue	Yes (Move Limits)	Low
Pet City	Facebook	Build, Visit Friends	Yes (Energy System)	Moderate

Table1 : A systematic analysis of 20 commercially successful mobile pet simulators

Design specifications: The HUD adopts a retro anime and candy color style (similar to muse dash). And it will include:

1. A fixed-room monitoring layout (w rooms) with a single active camera feed at a time
2. A scripted pet behavior timeline, randomly selecting events from a limited pool (e.g., knock over, chew object)
3. A limited-use intervention system (e.g., “throw toy,” “activate sound”), with cooldowns and delayed effect
4. A UI overlay that includes both surveillance controls and passive background tasks (e.g., event planning pop-ups)

These specifications help translate abstract design theories into executable gameplay elements suitable for testing.

Worldbuilding and Narrative Primer has a short opening comic sets the scene: the player is about to leave for work, connect to a surveillance system, and must balance daily and realtime pet care. This opening comic not only outlines the player's emotional path but also reinforces the theme of "limited control."

Optional Research: We plan to conduct informal player interviews and digital surveys to collect player feedback on game tension, interface clarity, and emotional reactions to missing interventions.

Design Artifact:

Phase I (Paper I):

Phase I (Paper I): Build and test a minimum viable prototype, focusing on the core loop:

1. Three-chamber monitoring system
2. Two pet behavior types (scripted)
3. Three intervention methods, with cooldowns and delays
4. Visual feedback by flashing the screen or shaking an object
5. Basic interruption task
6. UI (optional toggle) Pet status UI panel showing real-time values (emotion, hunger, destructive drive)

Phase II (post-MVP/Thesis II):

Extend pet behavior logic using AI or behavior trees

Introducing dynamic task interruptions (e.g., emergency calls, scheduling requests)

Add a scoring or feedback system to reflect player success/failure rates

Develop a performance summary screen ("end of day" report)

Pre-Producti

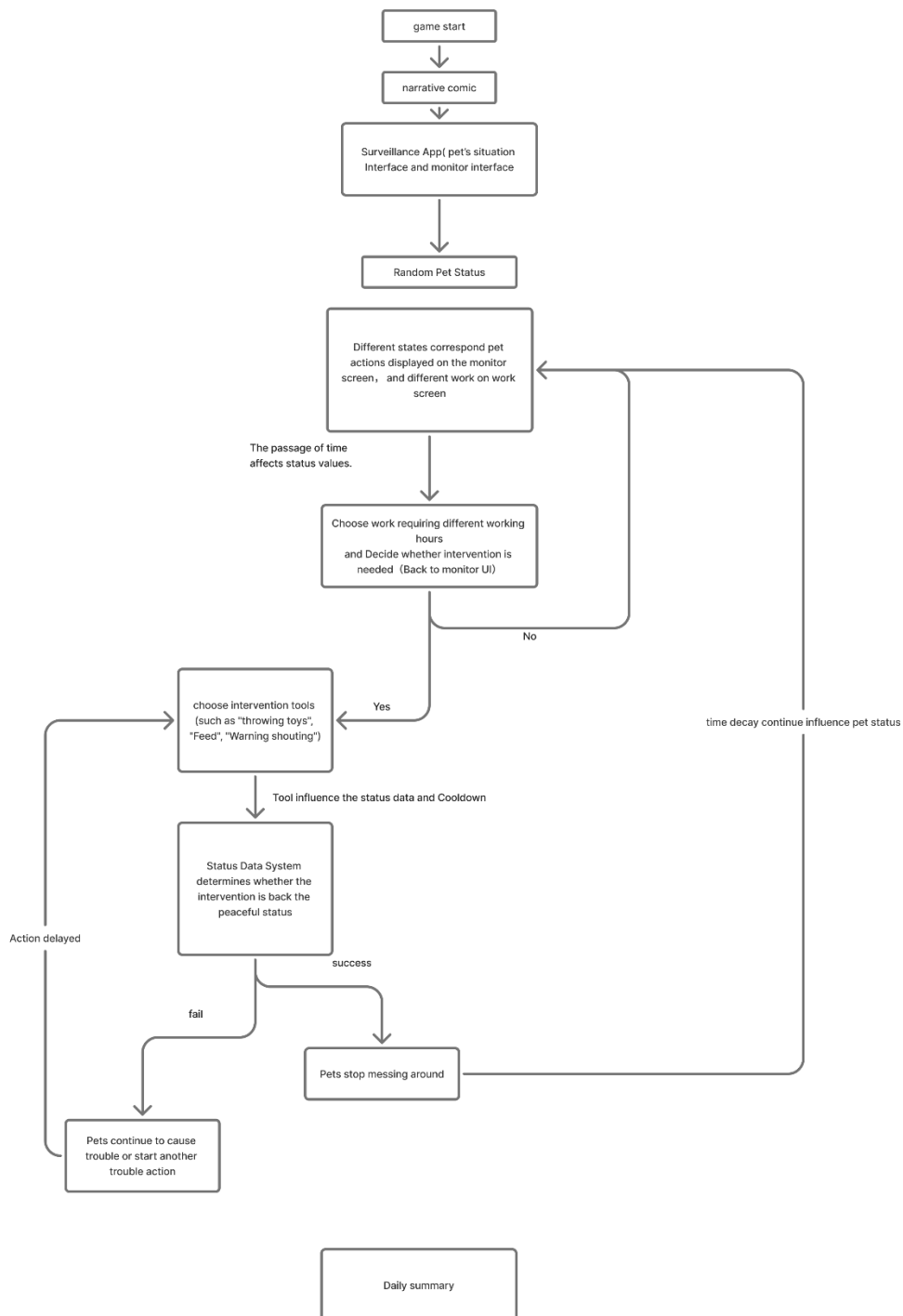


Figure2: Core Loop

UI framework:

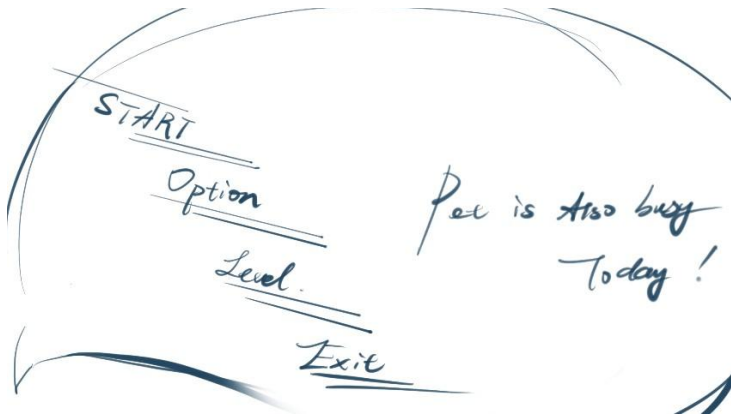


Figure3: Sketch of Main Menu

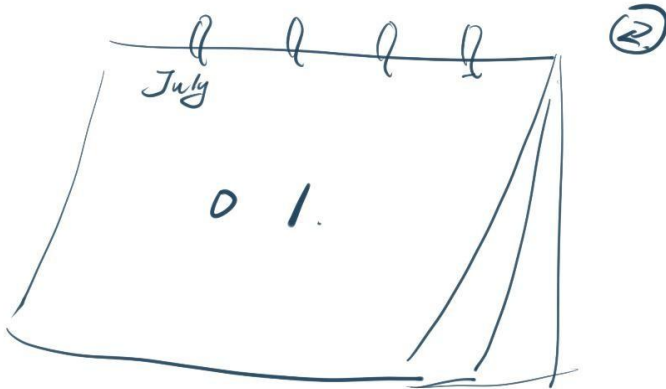


Figure4: Sketch of Level Option

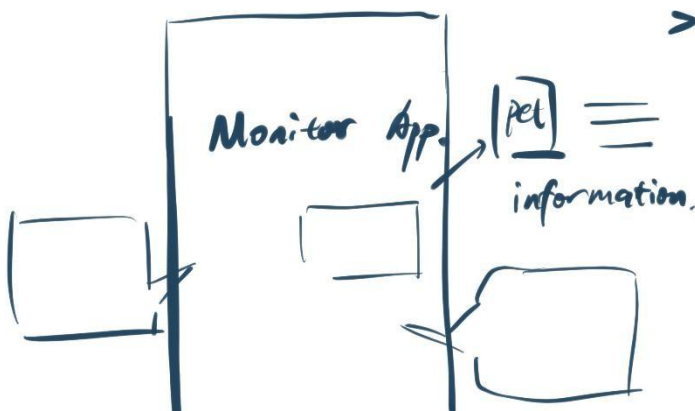


Figure5: Pet Information Menu:



Figure6: Pet Monitor Information

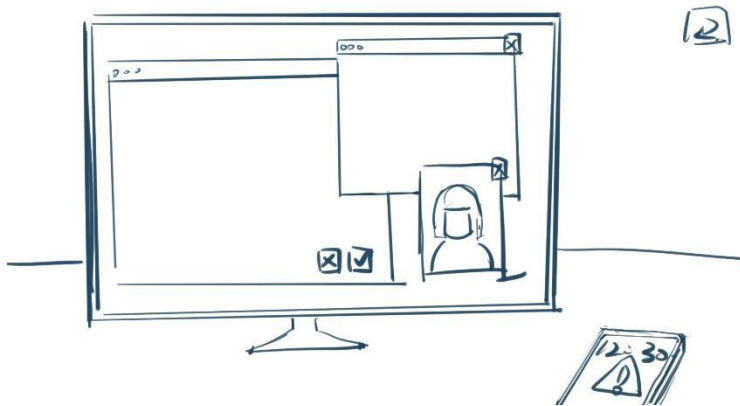


Figure7: Working Interface (can switch to monitor interface by clicking the phone)

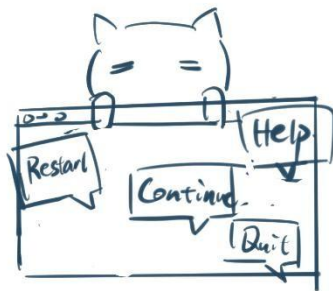


Figure8: Pause Menu

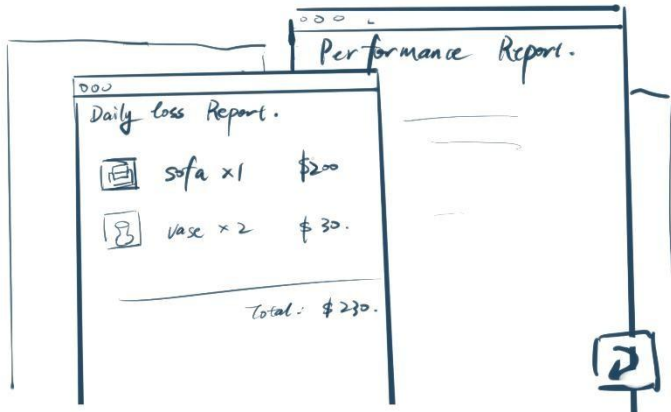


Figure 9: Summary Interface

Introduction.



My daily life is
like this ...



Figure10: Narrative Comic Storyboard (Introduce the background)

Development Team:

This prototype is being developed primarily by Zhiyun Yu, who is responsible for all gameplay systems, UI logic, interaction design, concept design and scripting in Unreal Engine 5.

Visual support (such as Pet design, environment design and character design) will be provided by Wanzhen Mei, a collaborator focused on post-production art assets. All core development and system logic are handled independently by the primary researcher.

Visualization:

Simple sketch collage (figure11, 2025) of general visual and color style, which will be dominated by candy colors because the overall game theme's emotional tone is playful to show the cute emotional connection between pets and people.

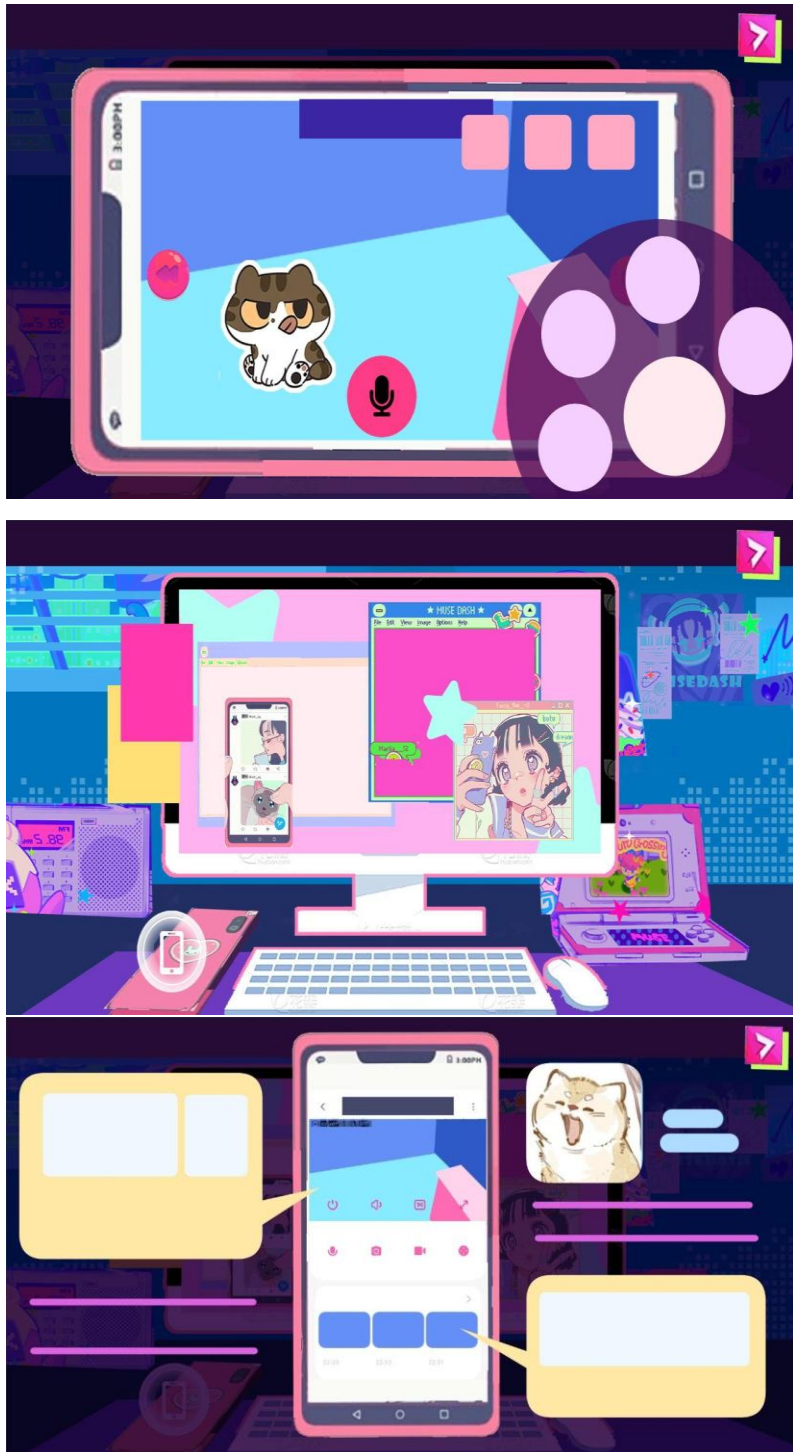


Figure11: visual style confirms

Development

Prototype Overview

The main goal of the prototype is system testing. Built using Unreal Engine, the prototype is designed to evaluate the core UI system and the functional relationship between pet behavior states and player actions. Specifically, it explores how players respond to the dynamic changes of three interrelated pet status values: happiness, hunger, and destructiveness. Players are given three behavioral intervention tools - feeding, throwing toys, and shouting - and need to choose the most appropriate action based on the pet's current state.

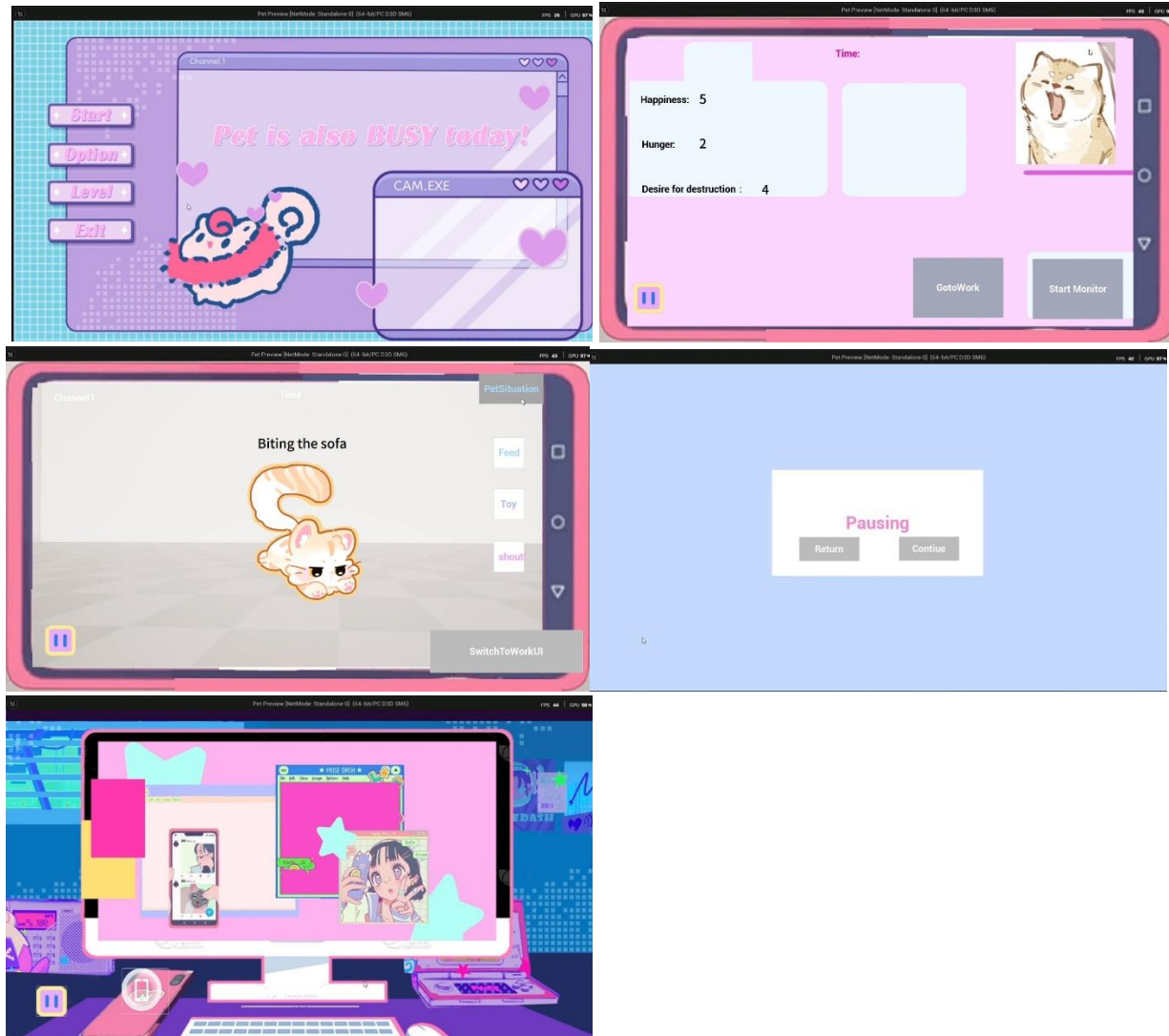


Figure12:Screenshot from paper Prototype

A key design principle behind the prototype is that the three values are initially random, but the three status values do not exist independently; they dynamically interact and influence each other. For example, feeding will reduce hunger, and when hunger reaches a certain threshold, it will increase the pet's happiness. Conversely, if the pet is not fed for a certain period and the hunger value rises to a certain value, its happiness value will decrease, while also increasing its destructiveness value. Similarly, shouting, while it can immediately suppress destructive behavior, may significantly reduce happiness,

thus triggering further negative reactions. This interdependency is the core of testing, requiring players to observe, interpret, and balance multiple constantly changing values to make the best decision.

The purpose of this prototype was therefore twofold: to assess whether the player could effectively and intuitively manipulate the user interface, and to determine whether the game features worked and successfully created a sense of strategic tension. To succeed, players must explore how their interventions affect the logic of interrelated state values and manage the balance of their pet's emotions and behaviors over time.

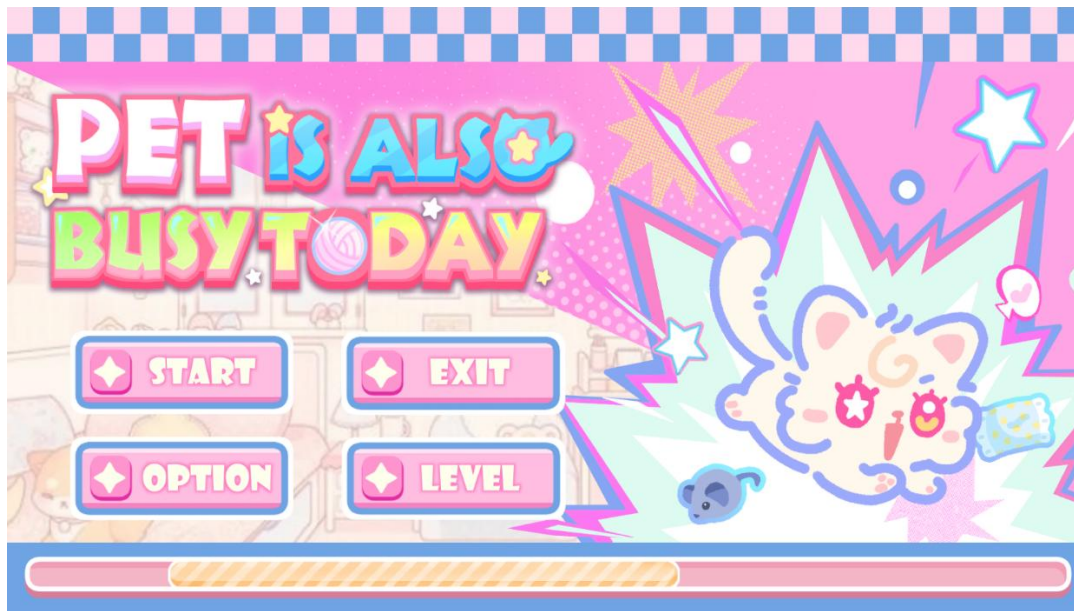
Second Prototype: Spatial Surveillance Restriction

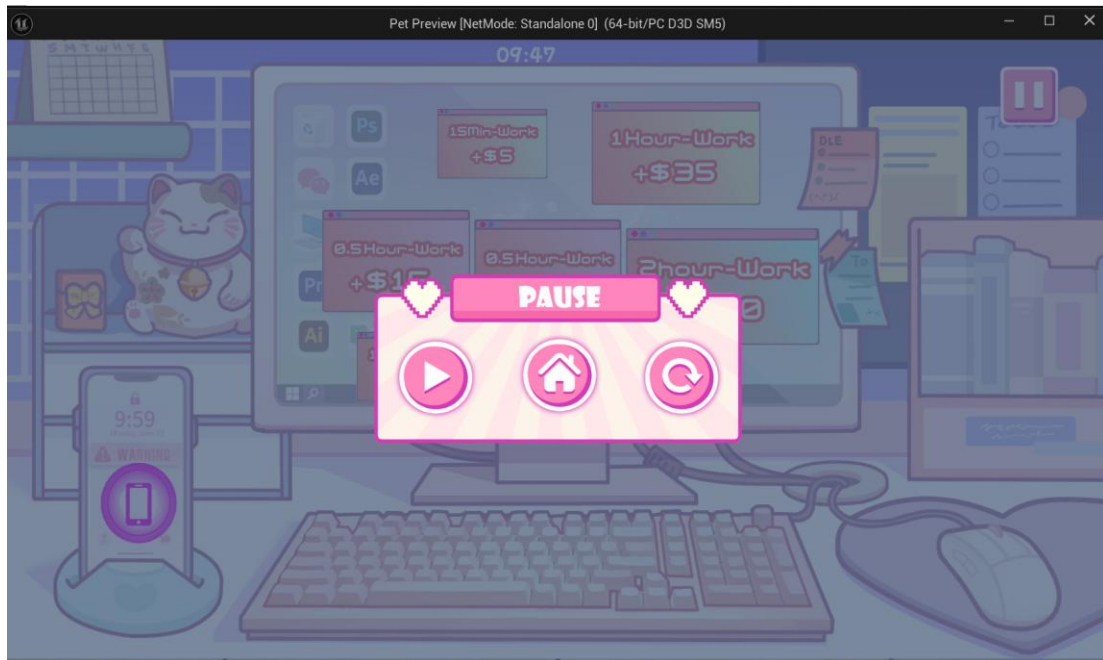
In this phase, I focused on implementing spatial restrictions within the monitoring system.

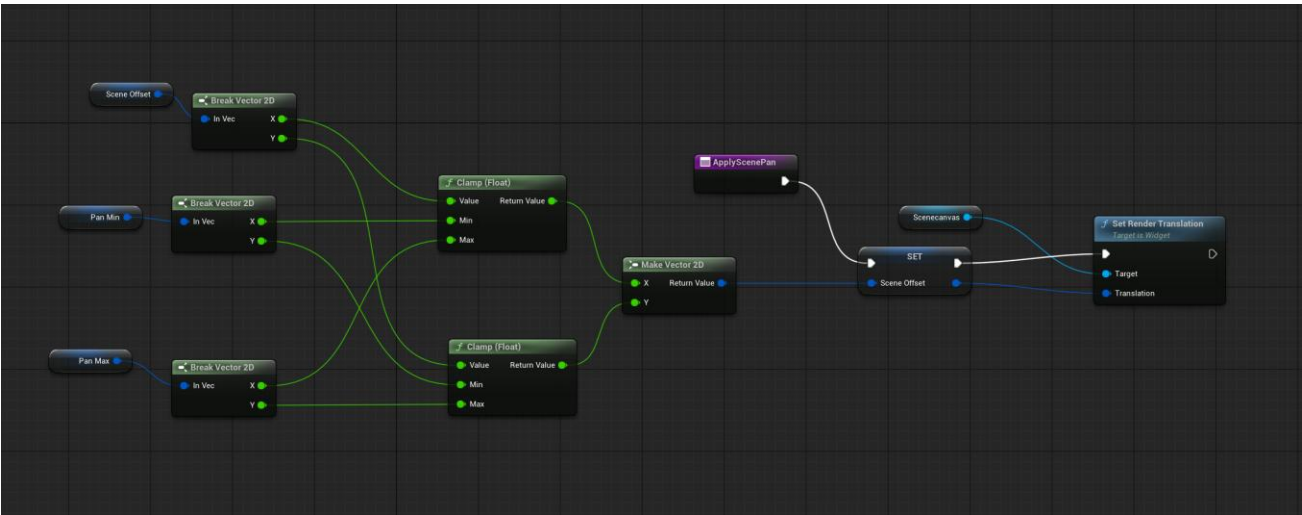
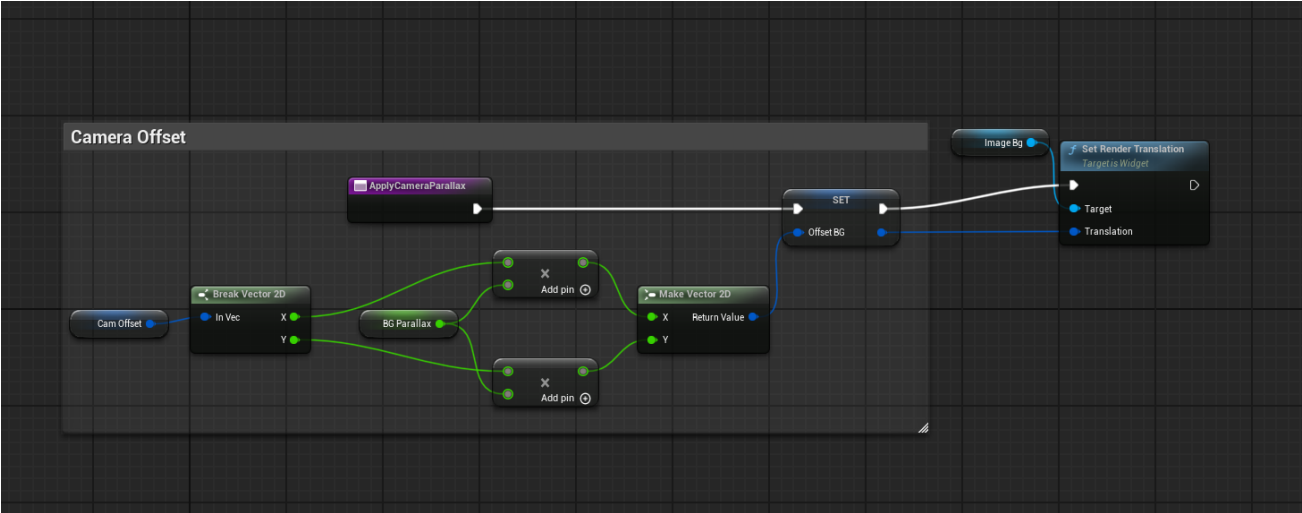
Players are required to actively move the camera (left, right, up, and down) to locate the pet, rather than seeing everything at once. In addition, they need to switch between the monitoring interface and the pet status panel to understand the pet's condition.

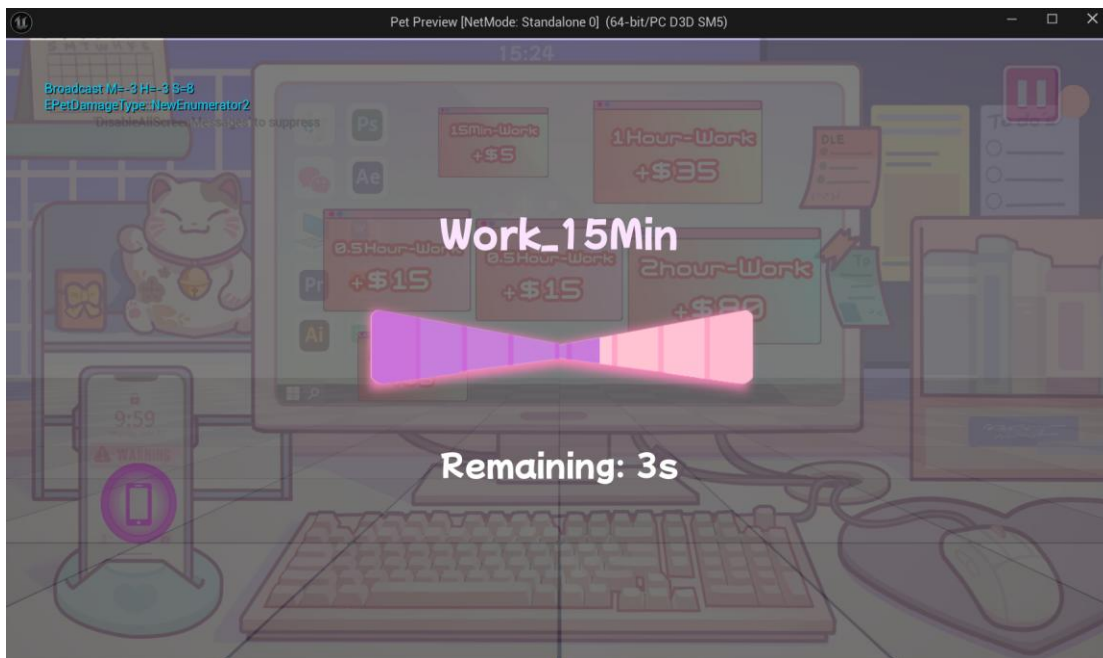
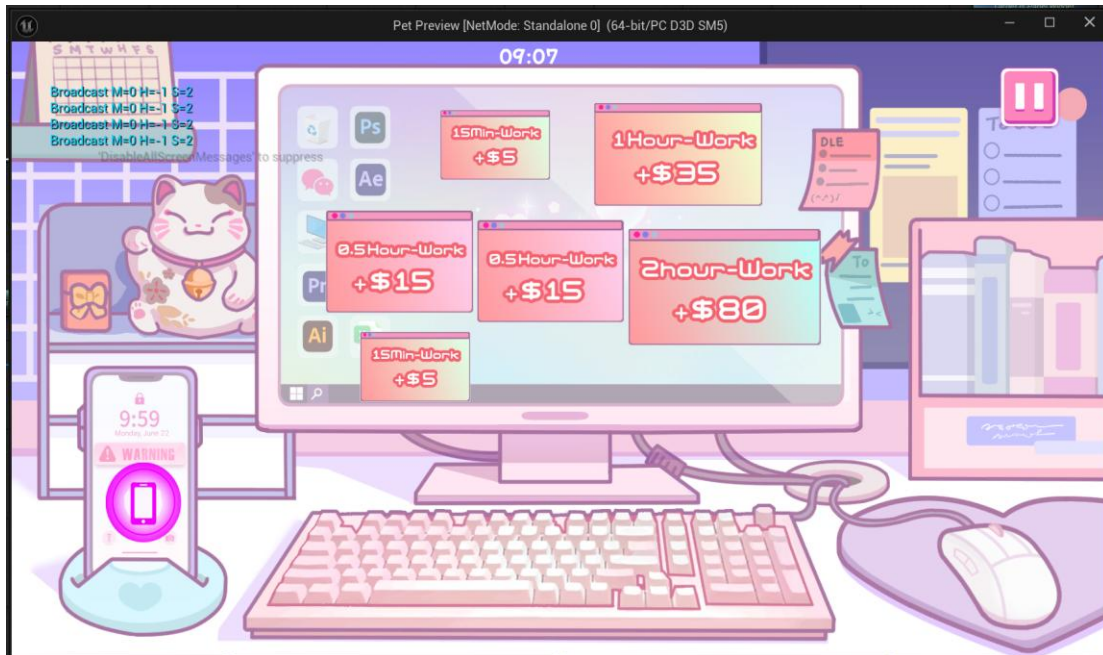
These mechanics mean that players cannot access all information at the same time. Instead, they have to search for the pet, check its status, and piece together what is happening.

Together, these interactions create a spatial surveillance constraint, where limited visibility and fragmented information force players to actively gather information and make decisions.

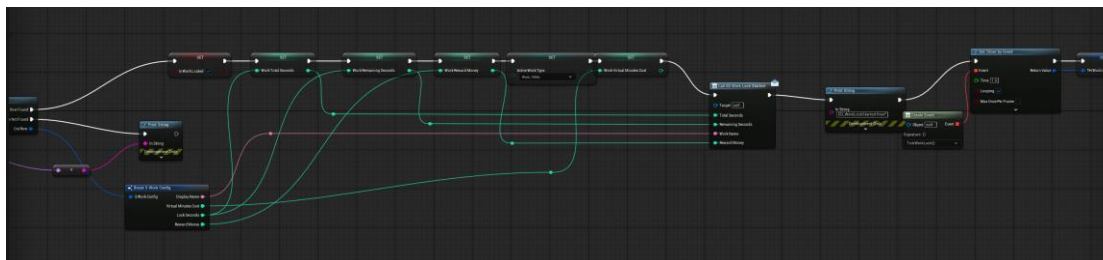


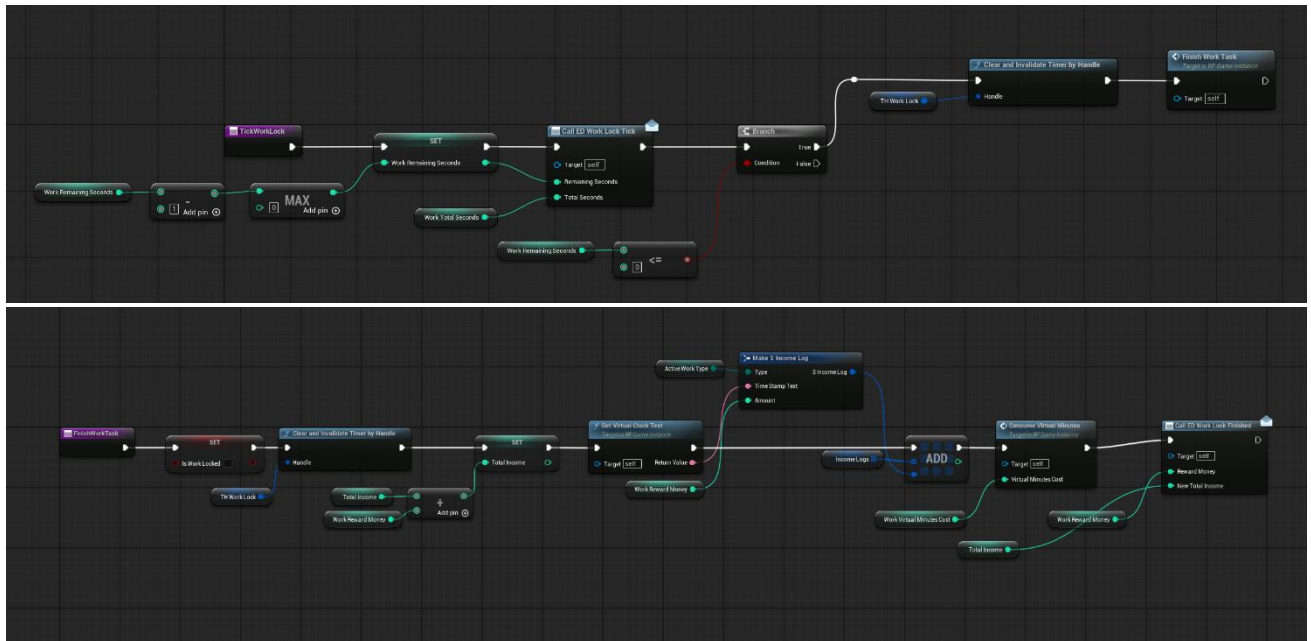




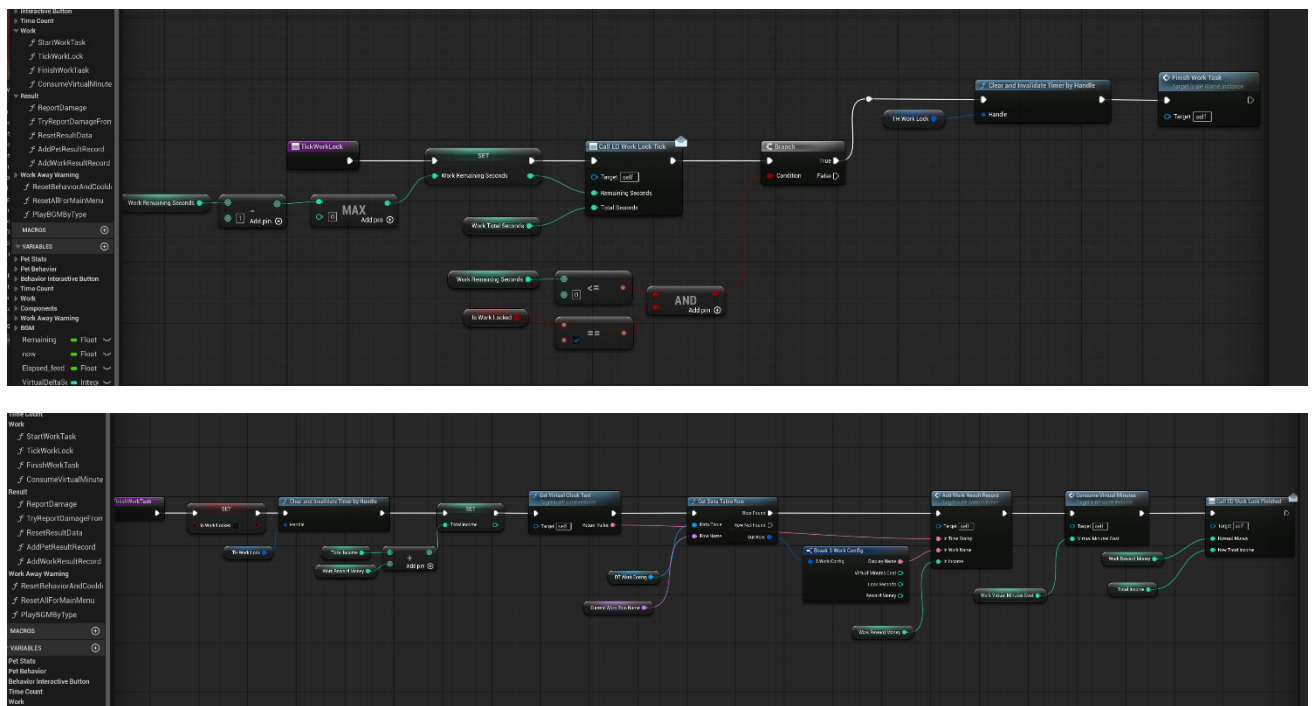


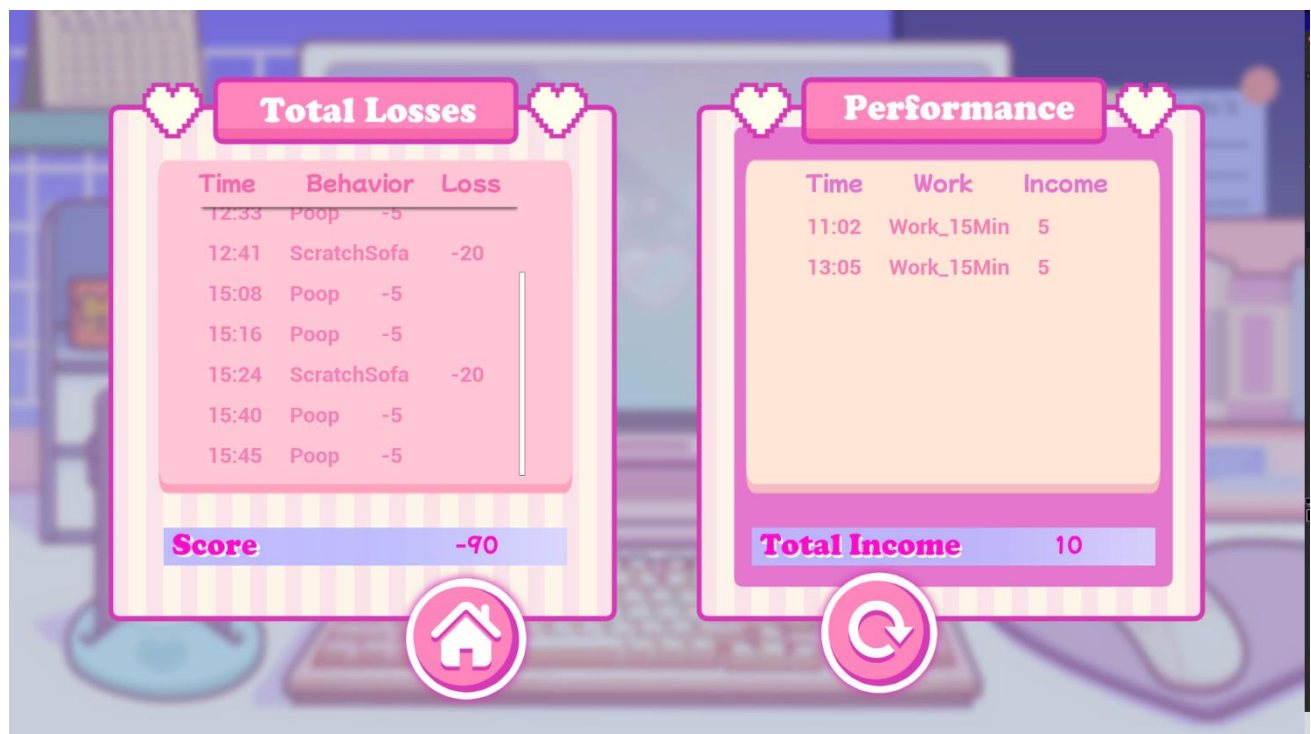
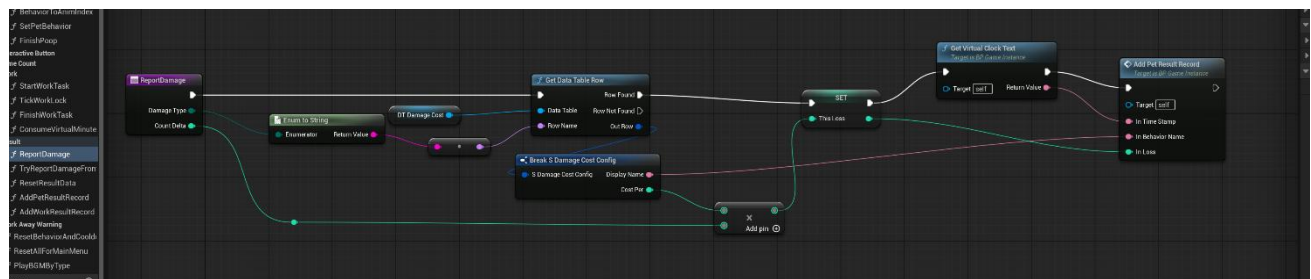
Add Work Menu Restriction





Add Time Record Stamp in Result Menu for easier monitoring and prepares for final subsequent data calculations:





Validation Methodology

Research Design Overview

This study employs a structured experimental design to assess the impact of surveillance-based constraints on player behavior and perceived psychological tension within a pet simulation framework. Based on the Theory of Constraints (TOC), the study sees surveillance access as the main bottleneck in the system, which limits the information that the player can get.

Rather than eliminating constraints, this research systematically alters the degree and type of restrictions to examine how players adjust their strategies in environments characterized by limited information and control. The validation process is divided into three steps, each of which adds a new type or level of constraint.

To account for differences in playstyle and decision-making tendencies between subjects, a within-subject design (Clark, Lawrence, Astley-Jones, & Gray, 2009) is used in all phases. In-game behavioral data are collected through.

Phase 1: Temporal Surveillance Restriction

The first phase examines how interruptions in monitoring access overtime influence player strategy and perceived tension.

In this condition, players are periodically forced into a “work” state, during which they lose access to both surveillance and intervention tools. This creates a temporal bottleneck, preventing continuous observation and requiring players to make decisions in advance.

The independent variable is the duration of monitoring absence:

- Short interruption (5 seconds)
- Medium interruption (10 seconds)
- Long interruption (20 seconds)

The primary behavioral metric is the **Pre-Event Monitoring Rate (PEMR)**:

$$PEMR = \frac{\text{Monitoring returns within pre-event window}}{\text{Total negative events}}$$

This metric measures predictive monitoring behavior by seeing if players go back to the monitoring interface just before something bad happens. A higher PEMR means that the person is better at planning ahead than reacting.

After each condition, people report their own tension on a Likert scale from 1 to 7.

This phase identifies temporal restriction as the principal system bottleneck, directly correlating with the Theory of Constraints by reducing the frequency of information access and necessitating proactive planning amidst uncertainty.

Phase 2: Spatial Surveillance Restriction

The second phase examines how reduced visual access to the environment affects monitoring behavior and strategic decision-making.

In this condition, players cannot view the entire environment simultaneously and must actively navigate the camera to locate the pet. This introduces fragmented information access, requiring players to reconstruct system states from partial observations.

The independent variable is the level of surveillance visibility:

- Full visibility (baseline condition)
- Reduced visibility (restricted condition)

The primary behavioral metric remains PEMR:

$$PEMR = \frac{\text{Predictive monitoring returns}}{\text{Total negative events}}$$

Utilizing the same metric as Phase 1 facilitates a direct comparison between temporal and spatial constraints, thereby isolating the impact of various information limitations on predictive monitoring behavior.

Self-reported tension is also collected using a Likert scale (1–7).

This phase extends the investigation by introducing spatial fragmentation, testing whether limiting information completeness (rather than access frequency) produces similar or distinct strategic adaptations.

Phase 3: Integrated Constraint Validation

The third phase examines how players respond to **combined spatial and temporal constraints**, representing a higher level of systemic pressure.

In this condition, spatial visibility is reduced while temporal interruptions are simultaneously introduced, limiting both **information access and control frequency**.

Two experimental conditions are tested:

- Temporal restriction only (baseline)
- Combined temporal + spatial restriction

The primary behavioral metric in this phase is the **Critical Miss Rate (CMR)**:

$$CMR = \frac{\text{Missed critical events}}{\text{Total critical events}}$$

An event is considered critical if it crosses a threshold that requires intervention within a certain time frame. CMR measures strategic failure when there is a lot of pressure on the players, which shows whether or not they can keep up effective monitoring and intervention behavior when there are more constraints.

Secondary measures include:

- Self-reported tension
- Monitoring interval variability (exploratory)

This phase tests how well player strategies hold up when there are more complicated constraints and whether combined restrictions cause players to stop being able to predict what will happen.

Appendix A: Post-Play Questionnaire

1. What is your age?

2. How often do you play video games?
3. Have you played virtual pet or life simulation games before?
4. Do you currently have a pet at home?
5. What is your MBTI personality type?
6. If you had to describe your game personality, which describes you best?
7. Were the intervention buttons responsive and understandable?
8. Did the cooldown/delay system for actions add meaningful challenge?
9. Did you feel any sense of psychological pressure or urgency while playing?
10. Did you find yourself planning, prioritizing, or predicting what the pet would do next?
11. Did the game make you feel emotionally engaged or empathic toward the pet?
12. Did you ever feel guilty or frustrated when you failed to stop the pet's actions?
13. When playing, did you empathize with real-life pet situations?
14. What did you find most enjoyable or unique about the game?
15. What would you improve or adjust?

(Figure11: First Prototype testing feedback questionnaire)

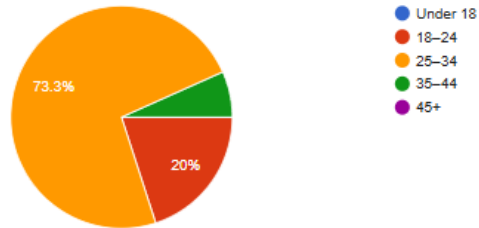
Survey Results and Analysis:

First and Second Demo feedback:

1.What is your age

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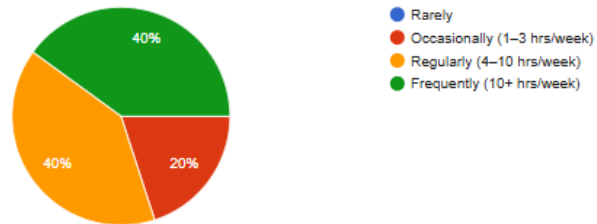
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2. How often do you play video games?

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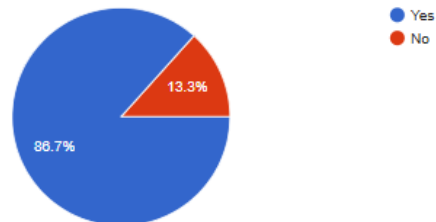
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3. Have you played virtual pet or life simulation games before?

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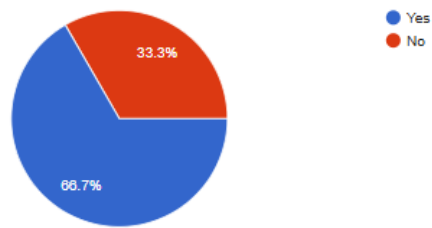
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4. Do you currently have a pet at home?

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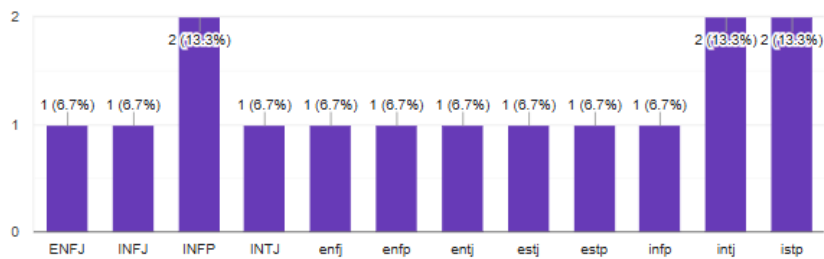
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5. What is your MBTI personality type?

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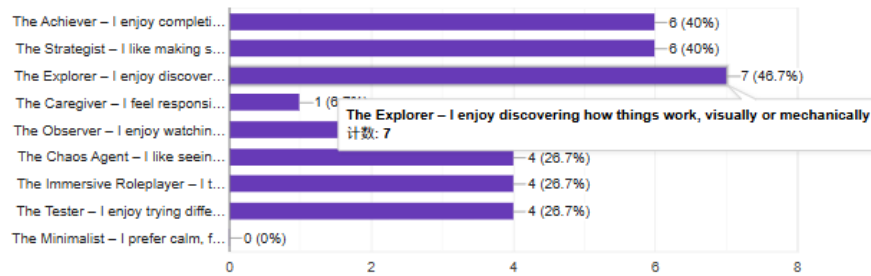
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6. If you had to describe your game personality, which describes you best?

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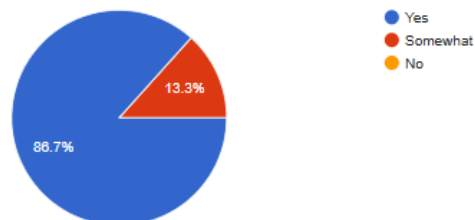
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7. Were the intervention buttons responsive and understandable?

(15 条回复)

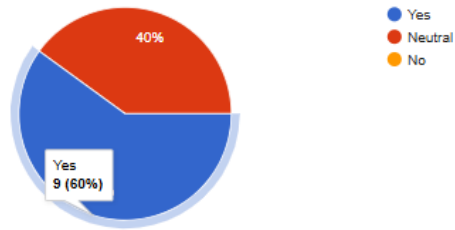
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8. Did the cooldown/delay system for actions add meaningful challenge?

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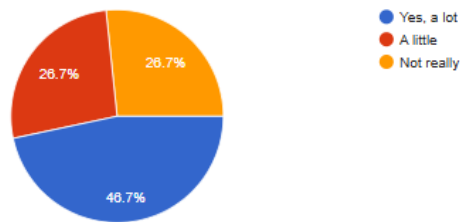
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9. Did you feel any sense of psychological pressure or urgency while playing?

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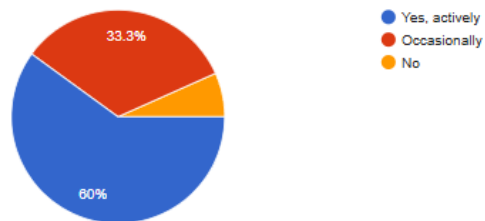
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10. Did you find yourself planning, prioritizing, or predicting what the pet would do next?

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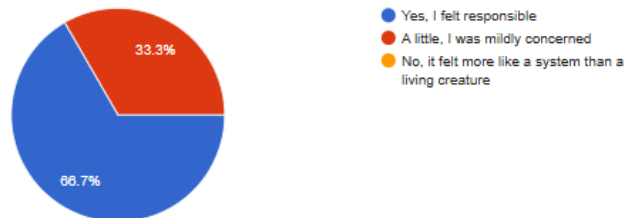
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11. Did the game make you feel emotionally engaged or empathic toward the pet?

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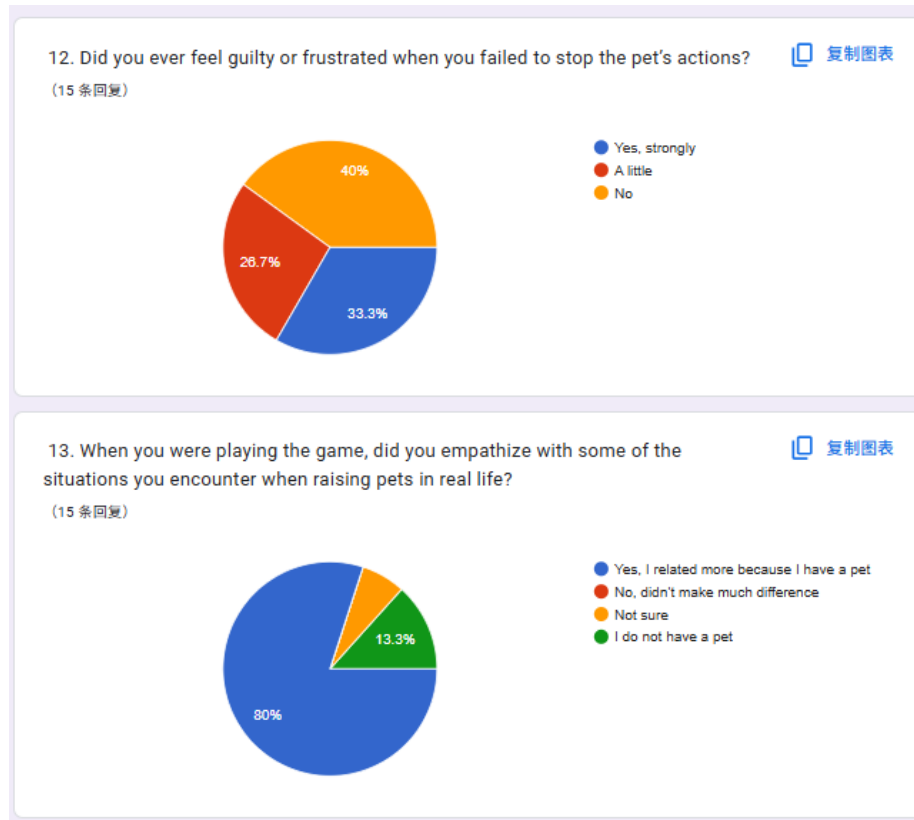


Figure12: Results of Prototype testing feedback questionnaire

Version 2 Test Feedback Interview

The second prototype (Version 2 Demo), five people took part in a one-on-one test round. The goal of this phase was to watch how players interacted with each other, how well they understood the system, and how much fun they had playing the game with the new design.

All players said they felt a noticeable amount of psychological tension while playing, especially in the middle of the game when they couldn't keep an eye on their pet all the time. All the participants felt this tension all the time.

Participant 2-1 noted that although the game created a sense of tension, the intensity was limited due to a lack of clear consequences: "I can feel some tension when I can't watch the pet, but it's not very strong because nothing really happens if I don't react immediately. There's no clear punishment."

Participant 2-2 expressed confusion about the consequences of pet behavior and the lack of feedback in the work system: "I didn't realize that the pet making trouble would cause a loss. And when I was forced back to the work screen, there was no penalty either, so I just went back to take care of the pet again."

Participant 2-3 commented on the lack of clarity in system priorities: "I wasn't sure which one I should focus on more, the pet or the work. It felt like both were important, but the game didn't really tell me what would happen if I ignored one."

Participant 2-4 highlighted the role of immediate visual feedback in shaping urgency: "I kept watching the pet because I could see what it was doing. And if I didn't pay attention, it would quickly go into a warning state and start doing something bad. That made it feel urgent, but I was mostly reacting to it instead of planning ahead."

Participant 2-5 also noted that the sense of urgency was primarily driven by visible warning cues: "It still felt urgent, because if I didn't watch the pet, it would quickly go into a warning state and start doing something bad. So I kept reacting to it."

Results:

The First and Second Demo:

From a functional perspective, the prototype performed well across all participants. All 15 testers were able to smoothly navigate between UI screens, including the main menu, pet status screen, and monitoring screen, without encountering any major errors or logic issues. Despite the use of placeholder visuals, the layout and flow were still considered intuitive. Perhaps due to the lack of finalization of the visual aesthetic, the framework appears relatively clear, with 13 of 15 participants (86.7%) rating the interface as clear or very clear. This also lays a solid foundation for future visual improvements.

In terms of strategic gameplay and tension, 12 participants (80%) reported experiencing a sense of decision-making pressure, particularly when deciding which tool (feeding, throwing toys, or yelling) to stop a pet's destructive behavior (such as chewing the couch or knocking over a cup). These behaviors are triggered when the "Destructive Desire" level reaches 6 or higher. Yelling reduces Destructive Desire but also reduces Happiness. A certain level of Happiness also increases Destructive Desire, further increasing the pressure of the decision. Four participants (26.7%) proactively mentioned comparing the effectiveness of different behaviors and observing how the pet's state (hungry, happy) affected destructive tendencies.

Several participants also emphasized confusion about the logic of the value system. Participant 1-3 specifically pointed out:

"I hesitated whether to shout or throw toys - both methods seemed to work, but I was not sure which one was more effective based on the pet's state."

However, qualitative feedback revealed a key conceptual problem: misunderstanding of state value. Two participants (Participant 1-1 and Participant 1-4) commented that the direction of happiness, hunger, and desire for destruction were inconsistent. For example:

Happiness (0-10) increases as your pet is happier.

Hunger (0-10) increases as your pet is hungrier, which confused some users because they thought higher values always reflected more positive states.

Participant 1-8 commented:

"At first I thought a hunger value of 0 meant my pet was hungry, but it was actually the opposite - it was full. This was a bit counterintuitive."

The Third Demo:

The results from the second prototype show that the changes made to the game successfully made players feel more urgency. Participants consistently reported experiencing tension, especially when the pet entered a warning state and seemed on the verge of exhibiting negative behavior. This shows that limiting surveillance and giving immediate visual feedback together created short-term psychological pressure.

However, the findings also reveal that this tension remained largely reactive rather than strategic. Players continued to prioritize immediate monitoring of the pet and responded to visible warning cues, rather than planning or allocating attention based on long-term considerations. A major problem found in Phase 2 is that people don't know what will happen if they don't follow the rules. Some people didn't fully understand that bad pet behavior could cost them money, and they didn't see the work system as having any real risk or reward. As a result, players didn't come up with a way to weigh the pros and cons of their choices.

In addition, the difference in how engaged people were with the interfaces had a big effect on their behavior. People found the monitoring system, which had animation and warning feedback, to be more interesting than the work system. This made players keep going back to monitoring, which made the gameplay loop more reactive.

Overall, Phase 2 did a good job of creating tension through limits and feedback, but it didn't fully achieve its goal of encouraging strategic decision-making.

Conclusion

The system is clearly getting better through the polish. Phase 1 found important problems with the system's clarity, especially when it came to understanding value structures. This made it harder for players to think strategically. In response, Phase 2 made changes to the structure, such as a new value system, a clear limit on surveillance, and an initial economic layer.

These changes worked to make players feel more tense and urgent while they played. But the results from Phase 2 show that the system still doesn't turn this tension into strategic behavior. Players still depend on quick reactions based on what they see right away, instead of making long-term plans or decisions.

To enhance the validity of the experimental framework in Phase 3, additional refinements are required. First, the link between what players do and what happens as a result needs to be made clearer, especially by making economic loss more visible and immediate. Second, the work system should be changed to make

meaningful trade-offs possible, making sure that decisions about how to spend time have clear strategic effects. Third, onboarding and system guidance should be better so that players can better understand how different factors affect each other and the results.

It's important to make these changes so that constraint functions as a source of tension and a way to measure strategic behavior, which will help the accuracy and reliability of later validation phases.

Limitations and Future Directions

Limitations:

First, the sample size was small and subject to survivorship bias. The 15 test participants were pre-selected, relatively close in age, and mostly pet owners. This may have affected emotional engagement and narrowed the diversity of strategic behaviors. Second, due to time constraints, the prototype's art assets were only basic frameworks, which limited participant feedback on aesthetics and emotional tone. Furthermore, due to the ambiguity of the status values, some participants' understanding of values like "hunger" and "happiness" deviated from their own, leading to discrepancies in their understanding of the game. This suggests the need for clearer UI metaphors or a consistent scaling approach.

Future Directions:

In the future, we will conduct more extensive and diverse user testing, including participants with different gaming personalities, MBTI traits, and cultural backgrounds. Secondly, we will upgrade the behavior logic from random scripted behaviors to finite state machines or lightweight AI to increase the variability and realism of pet behavior. Thirdly, we will add more art and animations and further optimize the user interface to enhance visual aesthetics. In addition, the addition of long-form gameplay and additional stress factors, the addition of overlapping work functions as external interference in the future, and time-based events will further enhance the sense of pressure and realism.

List of Figures and Tables

Figures:

Figure 1. From “MDA: A Formal Approach to Game Design and Game Research” by R. Hunicke, M. LeBlanc, and R. Zubek, 2004, “Proceedings of the AAAI Workshop on Challenges in Game AI”. Copyright 2004 by the authors.

Figure 2. Core loop

Figure 3. Sketch of Main Menu

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Tables:

Table1: A systematic analysis of 20 commercially successful mobile pet simulators

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Figma: Core Loop, Timeline Gantt Chart

ChatGPT: Assist in Translation, Grammar Checking

Google Form: Testing Feedback

Photoshop: Concept Design, Before-Game Comic, Pet and Character

Adobe Illustrator: UI

After Effects: Animation

Maya, Substance 3D Painter: Environment Model